Experiment 17:

Implement Mobile Price Prediction using appropriate machine learning algorithm

Python Code:

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
X = np.array([
    [2, 16, 5, 3000],
    [3, 32, 8, 4000],
    [4, 64, 12, 5000],
    [4, 128, 16, 6000],
    [6, 128, 20, 7000],
    [6, 256, 24, 8000],
    [8, 256, 32, 10000],
    [8, 512, 48, 12000],
    [12, 512, 64, 15000],
    [12, 1024, 108, 20000]
])
y = np.array([0, 0, 1, 1, 1, 2, 2, 2, 3, 3])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
new_mobile = np.array([[8, 256, 32, 10000]])
predicted_class = model.predict(new_mobile)
print("Actual values:", y_test)
print("Predicted values:", y_pred)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("New mobile specifications:", new_mobile[0])
print("Predicted price range:", predicted_class[0])
```

Output:

```
Actual values: [3 0]
Predicted values: [2 0]
Accuracy: 0.5
New mobile specifications: [    8   256    32 10000]
Predicted price range: 2
```